Python: The Easy Way

Lab 1

Python Installation

Windows

1 Download the convenient Python version from the official website:

```
Website: https://www.python.org/downloads/
```

2 Run the executable file downloaded and follow the instructions

3 Add Python to Path (Environment) Variables to be global command:

```
You will find The steps screenshots here:
http://www.howtogeek.com/197947/how-to-install-python-on-windows/
```

Linux or OS X

In most of Linux distributions, you will find python installed

```
python                          #for python 2

python3                         #for python 3
```

On Mac OS X

```
brew install python                     #for python 2

brew install python3                    #for python 3
```

Lab 1 Assignments

Vowels Removal

Write a program that remove all vowels from the input word and generate a brief version of it.

Input

String

"mobile"

Output

String

"mbl"

Bonus: You can make a function like replace Fn. but takes list of strings instead of one string.

Character Locator

Input

String

"This is javaScript"

Letter

"i"

Output

List

[2, 5, 15]

Multiplication Table

Write a program that generate a multiplication table from 1 to the number passed

Input

number

3

Output

List of Lists

[[1], [2, 4], [3, 6, 9]]

Calculate Area

Write a function that calculate the area of these shapes:

triangle = (0.5 * base * height), rectangle = (width * height), Circle= (PI * radius ^ 2)

Input

	string	number	number
Triangle	"t"	10	7
	string	number	number
Rectangle	"r"	10	7
	string	number	
Square	"r"	10	
	string	number	
Circle	"c"	10	

The Dictionary

Write a program convert a list of names into sorted dictionary which key is the Alpha. and value is a list of names corresponding to this alpha.

Input

List

[“ahmed”, “fatma”, “Ibrahim”]

Output

Dictionary

{a:[“ahmed”], f:[“fatma”], i:[“Ibrahim”]}

LAB +50 pts.

Mario Pyramid

Write a program that build a Mario pyramid like below:

Input

number

4

Output

String

```
   *
  **
 ***
****
```

Thank You